3.5.4

a.

Sum of digits is 9 means that $9 \mid n$

$\frac{9999-9}{9} + 1 = \boxed{1111}$

b.

Three numbers add up to 9 (1-1000, but 1000 doesn't work so its just 1-999), $\boxed{\frac{(9+2)!}{2!9!}}$ ways to have numbers add up to 9

The numbers can be zero, since $0001 = 1$ is valid!

c.

Three numbers that add up to 18, so there must be $\boxed{\binom{18+2}{2}}$ ways

3.6.5

Multiple of 99, so multiple of 11 and 9 at the same time

Starts at 1089, goes up to 9999

$abcd$ is divisible by 99 if $11 \mid (a - b + c - d)$ and $9 \mid (a + b + c + d)$

$x = a + c$, and $y = b + d$. Then $11 \mid (x - y)$, and $9 \mid (x + y)$

$x - y = \{0, 11\}$ - maximum since there's 4 digits only

$x + y = \{9, 18, 27\}$

$2x = \{9, 18, 27, 20\}$

$2y = \{9, 18, 27, 7, 16\}$

Actually, the possible values for $2x$ and $2y$ cannot be odd!

$2x = \{18, 20\}$

$2y = \{18, 16\}$

$x = \{9, 10\}$

$y = \{9, 8\}$

Checking against the initial conditions for $x - y$ and $x + y$, $x = 9$ and $y = 9$ otherwise it won't work

How do I solve this without bashing...

$9 = \{(1, 8), (2, 7), (3, 6), (4, 5)\}$ -> reverse these too

So we have a set of 8 total ways to make 9, and pick two distinct (so choose without replacement).

$\boxed{\binom{8}{2}}$